



# AMP: Arc Multi-Proposer Protocol

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Circle's open Layer-1 blockchain  
purpose-built for institutional finance

# Arc



## Battle-Tested Consensus Layer

**Malachite:** Tendermint  
implementation in Rust



## Sub-second deterministic finality

Settlement-grade  
operation, institutional  
network operators



## Smart Contracts with Predictable Costs

EVM-compatible with  
stablecoins as gas for  
low, fiat-based fees

# The single proposer bottleneck

Most blockchains delegate block construction and propagation to a single validator per height:

$$\text{throughput} \sim \frac{\text{block size}}{\text{block latency}}$$

Block size and latency **compete**: larger blocks take longer to propagate, thus increasing block latency.

Motivation: performance

# The block assembler's discretion

The block assembler concentrates powers:

- **Inclusion:** exclude transactions → censorship
- **Ordering:** reorder transactions → MEV extraction

According to the Bank for International Settlements (BIS), validator-driven transaction censoring and reordering are “activities that would be **illegal** in traditional markets”.

Motivation: control

# Building Block: Vote Extensions

Validators attach **arbitrary, signed** data directly to their consensus votes.

Consensus decisions now contain **votes** plus their **extensions**.

Already used in **Arc**, **Cosmos**, and **Polygon**.\*

Easily generalizable to other consensus protocols.

\* Multimmit – brand new paper from Commonware uses Vote Extensions

# Building Block: Proposers

Participate in **constructing** blocks

- Do not need to be validators

Can be **applications** (DEXs, lending, etc) who want to control their own transaction flow

# Design Overview

**Proposer**



*payload*



**Validators**



*attestation (VE)* =  
set of payload IDs



**Block Assembler**



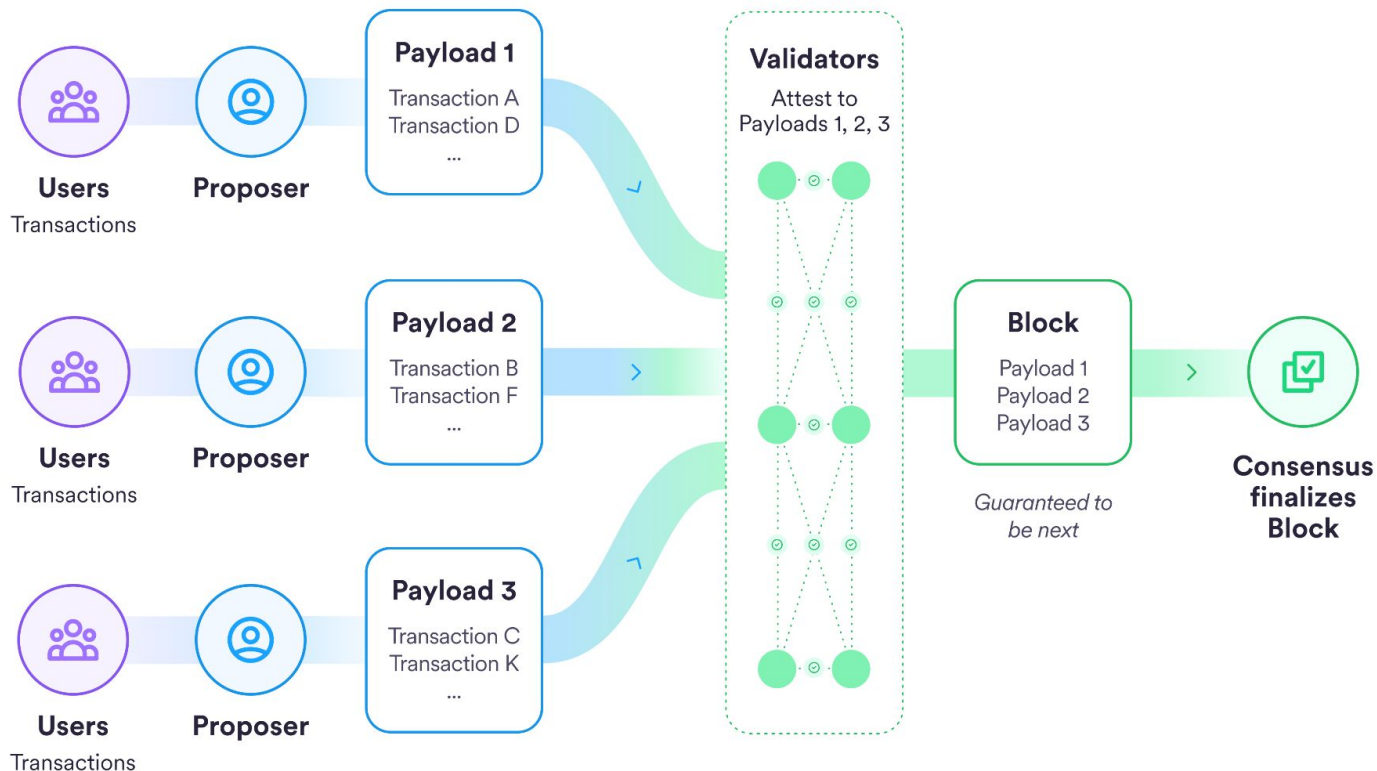
Consensus on a set  
of *attestations (VE)*



**Validators**



Finalize a block =  
*attested payloads*



# Properties: Bounded Inclusion & Fixed Ordering

The values proposed and decided via consensus are **sets of attestations**:

- The value proposed at height  $h + 1$  is the **commit certificate** of height  $h$

**Bounded Inclusion:** a payload **attested** by **many** ( $N - F$ ) validators at height  $h$  is **finalized** at  $h + 1$

The value finalized every height by **AMP** is a **block**, composed of **ordered** payloads

**Fixed Ordering:** transaction execution is decided by a **deterministic, public** rule

**Bonus:** Decoupling dissemination from ordering—**eliminates mempool** and **increases throughput**

# PBS vs AMP

|                      | PBS                                   | AMP                                                |
|----------------------|---------------------------------------|----------------------------------------------------|
| Transaction Flow     | User → Searcher → Builder → Validator | User → Proposer → Validator                        |
| Inclusion Likelihood | Based on relative fees.               | <b>Guaranteed</b> after N-F attestations.          |
| Block Construction   | Single winning builder.               | Commit certificate of previous block– <b>MCP</b> . |
| Transaction Ordering | Builder discretion.                   | <b>Deterministic.</b>                              |

# Open Questions

- **Byzantine proposers:** payloads selectively disseminated, need for retransmissions
- **Duplicate transactions:** clients send the same transaction to multiple proposers
  - Censorship Resistance vs Throughput in Multi-Proposer BFT Protocols (<https://eprint.iacr.org/2026/126.pdf>)
- **Proposer incentives:** motivating apps and others to act as proposers
  - Escaping the Monopolist: An Economic Model of Censorship Resistance in Multi-Proposer Protocols

# Takeaways

**AMP: a multi-proposer layer atop unmodified Malachite** — not a new consensus protocol:

- **Dissemination decoupled from ordering:** proposers broadcast payloads in parallel, consensus on compact payload IDs
- **Proposer controlled flow:** no intermediaries between proposers and validators
- **Bounded inclusion:** payloads attested by **N - F** validators included in the **next block**



# AMP: Arc Multi-Proposer Protocol with Bounded Inclusion Guarantees

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# Constellation vs AMP

|                         | Constellation                                                           | AMP                                                                               |
|-------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Proposers</b>        | 16 proposers who are also active network validators.                    | <b>No fixed limit</b> on proposer count; not required to be validators.           |
| <b>Block Proposal</b>   | Wait to hear from half of attesting validators (sequential dependency). | <b>No waiting required</b> ; looks directly at the last decided block to propose. |
| <b>Block Content</b>    | Carry full transactions and payload data.                               | <b>Compact blocks</b> containing only small payload identifiers.                  |
| <b>Consensus Change</b> | None. Maintains standard consensus votes without modifications.         | Yes. Vote extensions <b>slightly increase</b> the size of consensus votes.        |
| <b>Trust Model</b>      | Includes active and automated misbehavior detection for proposers.      | <b>Honest assumption</b> for proposers.                                           |